

Ground Truth

$$y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \cdot x_1 x_2, \theta_{real} = [a, b, c, d] = [0.5, -2, 0.8, 1]$$

LLM-SR
(Baseline)



Base Fit

$$\theta_{base} = [0.9, 1, -0.2, 0.5]$$



Mixed Fit

a, b, d → linear parameters
c → nonlinear parameter

$$\theta_{mix} = [0.7, -3, 1, 0]$$



Turn 1: Add x_1 coefficient to the bias term.

$$y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \cdot x_1$$

$$\theta_{mix}^1 = [\dots, \dots, \dots, \dots] \rightarrow s_1$$

Turn 2: Further add x_2 coefficient.

$$y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \cdot x_1 x_2$$

$$\theta_{mix} = [0.5, -2, 0.8, 1] \rightarrow s_2$$

Scored-based grouping

$$s_0 \begin{cases} y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \\ y = a \cdot x_1 + b \cdot x_2 + c \text{ (shortest)} \\ y = a \cdot \sin(c \cdot x_2) + b \cdot x_2^2 + c \end{cases}$$

Semantic-based grouping

$$\begin{aligned} s_0 & y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \\ s_1 & y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \cdot x_1 \\ s_2 & y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d \cdot x_1 x_2 \end{aligned}$$

New Expression Skeleton

$$y = a \cdot x_1 + b \cdot \sin(c \cdot x_2) + d$$

STRIDE
(Ours)



Equation Generation

Equation Evaluation

Experience Management